

Longest Repeated Substring

Problem Statement

Longest Repeated Substring (LRS) is defined as: Given a string, we want to find the longest repeated non-overlapped substring. For example: if the input is BANANA, the possible LRS are AN (BAN ANA), and NA (BANA NA). ANA is NOT LRS, because the last 'A' is overlapped. (BANANA)

Input

There are multiple test cases. For each test case, there is a string whose length is n ($0 < n < 10^7$). All the characters are uppercase.

Input ends with EOF.

Output

For each case, output the **lexicographically smallest** longest repeated substring separated by a space. Print "No LRS is found." if there is no LRS found in the input.

Sample

Input	Output
BANANA	AN
GATTACA	A
GAGAGAG	GAG
GATTACAGATTACA	GATTACA
TGAC	No LRS is found.
TGTAC	T
TTGGAACC	A